

# Static Gesture Classification — Run Report

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## Data paths

|                       |                                                                        |
|-----------------------|------------------------------------------------------------------------|
| Training data path 1  | /home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 30 trials       |
| Training data path 2  | /home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 30 trials     |
| Training data path 3  | /home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 30 trials     |
| Training data path 4  | /home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 30 trials    |
| Training data path 5  | /home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 30 trials      |
| Training data path 6  | /home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static · 30 trials      |
| Training data path 7  | /home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 31 trials     |
| Training data path 8  | /home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials    |
| Training data path 9  | /home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 30 trials      |
| Training data path 10 | /home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 31 trials     |
| Training data path 11 | /home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 30 trials    |
| Training data path 12 | /home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 32 trials       |
| Training data path 13 | /home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 30 trials |
| Training data path 14 | /home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials       |
| Training data path 15 | /home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 30 trials     |
| Training data path 16 | /home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 30 trials    |
| Total training paths  | 16 · 454 trials total                                                  |
| External test root    | /home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static                   |
| Report output dir     | /home/jestin/ThesisRepo/ML/saved_reports/AugmentationExperiments       |

## Sensor selection

|                |                                                                                                     |
|----------------|-----------------------------------------------------------------------------------------------------|
| Hands          | left                                                                                                |
| Segments       | palm_mid, thumb, index, middle, ring, pinky, wrist                                                  |
| Modalities     | YPR, Accel, Flex                                                                                    |
| Sensor columns | 88 channels                                                                                         |
| Classes        | 6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman |

## Preprocessing, features & split

|                             |                                             |
|-----------------------------|---------------------------------------------|
| Resample steps              | 90                                          |
| Butterworth                 | off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz |
| Normalisation               | minmax                                      |
| Feature mode                | stats+fft → feature dim 4752                |
| Test size · seed · CV folds | 0.2 · random_state=42 · 10-fold stratified  |
| Sample counts               | train (post-aug)=1452 · test=91             |

## Augmentation (training only)

|                        |                        |
|------------------------|------------------------|
| Enabled                | yes                    |
| Copies per train trial | 3                      |
| Jitter                 | off · sigma=0.01       |
| Amplitude scale        | on · range=(0.9, 1.1)  |
| Time warp              | off · range=(0.9, 1.1) |

## Classifiers — cross-validation & hold-out test

| Algorithm     | CV mean (10-fold) | CV std | Hold-out test acc |
|---------------|-------------------|--------|-------------------|
| Random Forest | 0.9993            | 0.0021 | 0.9780            |

## Per-class hold-out F1

| Class              | Random Forest | Support |
|--------------------|---------------|---------|
| Double_Closed_Fist | 0.970         | 16      |
| Double_Okay        | 1.000         | 15      |
| Double_Open_Palm   | 1.000         | 15      |
| Double_Phone       | 0.968         | 15      |
| Double_Pistol      | 0.929         | 15      |
| Double_Spiderman   | 1.000         | 15      |
| macro avg          | 0.978         | 91      |

## External test (NewTestData)

| Algorithm     | External test acc |
|---------------|-------------------|
| Random Forest | 0.8000            |

## Per-class external accuracy

| Class              | Random Forest | Support |
|--------------------|---------------|---------|
| Double_Closed_Fist | 0.800         | 5       |

|                  |       |   |
|------------------|-------|---|
| Double_Okay      | 1.000 | 5 |
| Double_Open_Palm | 1.000 | 5 |
| Double_Phone     | 0.000 | 5 |
| Double_Pistol    | 1.000 | 5 |
| Double_Spiderman | 1.000 | 5 |